METASPLOITABLE 2 SECURITY ASSESSMENT

AI-Assisted Enumeration using Ollama and HexStrikeAI

Student Name	Md. Jahid Matubber
Target System	Metasploitable 2
Target IP	192.168.10.130
Kali Linux IP	192.168.10.129 (shown in generated report screenshot)
AI Integration	Ollama with llama3.2:1b via local API endpoint
AI / Tool Orchestration	HexStrikeAI with MCP server integration
Assessment Type	Educational vulnerability assessment and web enumeration
Evidence Source	User-provided jahid.pdf (9 pages of screenshots)

1. Executive Summary

This report documents a controlled security assessment of the Metasploitable 2 virtual machine at 192.168.10.130 from a Kali Linux environment. The supplied evidence shows connectivity testing, Nmap service discovery, HTTP response inspection, web content enumeration, SearchSploit lookups, and an AI-assisted workflow using Ollama and HexStrikeAI.

The screenshots show Ollama configured with model llama3.2:1b through the local API endpoint http://localhost:11434/v1, and HexStrikeAI interacting with MCP tools. Evidence also shows successful ping, exposed services, HTTP 200 responses from Apache and Tomcat, and discovered web paths. The final HexStrikeAI phase report records /dav/ directory listing and Tomcat release notes as observed findings, while phpinfo disclosure and CVE-2008-2938 are marked potential rather than confirmed.

No successful exploitation is demonstrated in the supplied PDF. SearchSploit results are reference matches only, and the final evidence explicitly notes that no new scan or exploit was executed by the reporting script. This report separates observed findings from potential risks and unverified leads.

2. Scope and Objectives

Target: Metasploitable 2 at 192.168.10.130. Kali Linux address shown in the final AI-generated report: 192.168.10.129.

- Verify network reachability.
- Identify exposed services and versions using Nmap.
- Inspect HTTP responses on Apache and Tomcat.
- Enumerate web directories and application paths.
- Search public exploit references for observed server software.
- Use Ollama through HexStrikeAI/MCP to assist with tool discovery, interpretation and reporting.

Authorization assumption: testing was performed against the user's own intentionally vulnerable lab VM.

3. Tools and Technologies Used

Tool / technology	Role	Evidence in PDF
Kali Linux	Testing workstation	Desktop and terminal throughout.
Metasploitable 2	Intentionally vulnerable target VM	Target console and target IP.
Ollama	Local model runtime	Pages 1–3 show invocation/configuration and local API endpoint.
llama3.2:1b	Selected local language model	Page 2 debug configuration shows model ollama:llama3.2:1b.
HexStrikeAI	AI-assisted workflow and orchestration	Pages 1–3 and 7–9.
MCP server	Tool integration interface	Pages 1–2 show loaded tools and MCP connection.
Nmap	Host and service discovery	Page 4.

curl	HTTP response-header inspection	Pages 4–5.
SearchSploit	Exploit-DB reference searches	Pages 6–8.
Directory enumeration / brute-force phase	Web path discovery	Phase checklist page 9 marks it DONE; paths summarized page 8.

4. Methodology

4.1 Ollama and HexStrikeAI Setup

The screenshots show HexStrikeAI with MCP server integration. The debug configuration on page 2 identifies the model as ollama:llama3.2:1b, provider API key as set, and provider URL as http://localhost:11434/v1. The system prompt requests the response “Hello”, and a later screenshot shows the model loaded and replying. An earlier interaction reports an “invalid model name” error; later configuration shows the selected model and subsequent response.

HexStrikeAI lists tools available from MCP servers and prompts the operator to use them only within an authorized Metasploitable 2 lab. This demonstrates AI/tool integration and workflow preparation; it does not prove every listed tool was executed.

4.2 Connectivity Verification

Page 4 shows `ping -c 4 192.168.10.130`: four packets transmitted, four received, 0% packet loss, and average round-trip time approximately 1.046 ms. This confirms basic IP reachability at the time of testing.

4.3 Port and Service Discovery

The Nmap output on page 4 reports the target as up and lists multiple open TCP ports. The final HexStrikeAI report on page 8 summarizes these selected services:

Port	Service / product	Observed version
21/tcp	FTP	vsftpd 2.3.4
22/tcp	SSH	OpenSSH 4.7p1
80/tcp	HTTP	Apache 2.2.8 (Ubuntu)
3306/tcp	MySQL	MySQL 5.0.51a
8180/tcp	HTTP / Tomcat	Apache Tomcat 5.5

The detailed Nmap screenshot also shows Telnet, SMTP, DNS, SMB/NetBIOS, RPC, Java RMI, NFS, ProFTPD, distccd, PostgreSQL, VNC, X11, IRC and AJP13. The table follows the condensed service summary from the AI-generated report.

4.4 HTTP Response Inspection

curl checks on pages 4–5 report HTTP/1.1 200 OK for http://192.168.10.130/ and http://192.168.10.130:8180/. The Apache response includes Server: Apache/2.2.8 (Ubuntu) DAV/2 and X-Powered-By: PHP/5.2.4-2ubuntu5.10. The Tomcat response includes Server: Apache-Coyote/1.1.

4.5 AI-Assisted Web Security Review

A HexStrikeAI prompt on page 5 requests analysis of the HTTP service and raises considerations such as outdated Apache/PHP components, potentially exposed administrative paths and HTTP security practices. Pages 6–7 show HTTP request text being submitted for analysis. These are AI-assisted review steps; they do not independently validate every suggested weakness.

4.6 SearchSploit Review

SearchSploit results on pages 6–7 show exploit-reference matches for Apache 2.2.8 and Apache Tomcat 5.5. The page 9 report states that searches were completed and that search results alone do not confirm a vulnerability. No exploit execution is shown.

5. Results and Findings

ID	Finding	Evidence / status	Risk interpretation
F-01	Target reachable	Confirmed: ping 4/4 replies, 0% loss (p.4).	Informational.
F-02	Multiple exposed services	Confirmed by Nmap (p.4).	High attack-surface exposure; services need individual review.
F-03	Apache HTTP responds	HTTP 200; Apache 2.2.8 (Ubuntu), DAV/2 (pp.4–5).	Outdated version is a potential risk; banner alone does not prove exploitability.
F-04	Tomcat responds	HTTP 200 on 8180; Apache-Coyote/1.1 (p.5).	Reachable application service; configuration needs review.
F-05	Web paths discovered	/dav/, /test/, /phpinfo.php, /phpMyAdmin/, /twiki/, /manager/html, /host-manager/ (p.8).	Sensitive endpoints require access-control review.
F-06	Directory listing at /dav/	Marked Confirmed; response showed “Index of /dav” (p.9).	Information disclosure.
F-07	Tomcat release notes accessible	/RELEASE-NOTES.txt returned HTTP 200 (p.9).	May disclose product/version information.
F-08	PHP information disclosure	Marked Potential at /phpinfo.php; response body needs review (p.9).	Not confirmed.
F-09	CVE-2008-2938 / Tomcat 5.5	Marked Potential; exact version/configuration not confirmed (p.9).	Unverified vulnerability lead.

6. AI-Generated Phase Report Summary

The terminal report on pages 8–9 is titled "HexStrikeAI — Phase 1 & Phase 2 Report". It records target/Kali IPs, successful ping, selected services, web enumeration results, findings, SearchSploit completion and a checklist.

Phase / checklist item	Status shown
Phase 1 — Nmap reconnaissance	DONE
Phase 2 — Web enumeration	DONE
Directory brute-force	DONE
Known CVE verification	PARTIAL

The source report states that it summarizes previously collected evidence and that no new scan or exploit was executed by the script. It instructs the reader to verify findings before marking them confirmed.

7. Recommendations

1. Keep Metasploitable 2 isolated in a host-only or otherwise controlled lab network.
2. Review open ports and disable services not required for the exercise.
3. Upgrade legacy Apache, PHP, Tomcat, database and remote-access components in real deployments.
4. Disable directory listing on /dav/ and restrict WebDAV access.
5. Remove or restrict phpinfo.php after diagnostics.
6. Restrict /manager/html and /host-manager/ to trusted administrative access with strong authentication.
7. Validate Tomcat version and configuration before treating CVE-2008-2938 as applicable.
8. Treat SearchSploit matches as leads only; confirm build, configuration and applicability.
9. Preserve raw scan outputs, timestamps, commands and screenshots for reproducibility.
10. Manually review AI-generated suggestions and distinguish tool output from model interpretation.

8. Limitations

- Based on screenshots in the supplied nine-page PDF; raw scan exports and packet captures were not supplied.
- Some terminal text is small or partially obscured; only readable details are reported.
- Evidence confirms connectivity, service responses, selected web paths and /dav/ listing, but not successful exploitation.
- SearchSploit matches do not prove exploitability.
- CVE-2008-2938 is marked potential; exact version and configuration applicability are not confirmed.
- Ollama configuration and model interaction are visible, but a complete prompt/response export is not provided.

9. Conclusion

The evidence demonstrates an AI-assisted assessment workflow in which Ollama with llama3.2:1b was configured through a local API endpoint and used with HexStrikeAI/MCP for tool discovery, interpretation and reporting. Network testing confirmed the Metasploitable 2 host was reachable and

exposed multiple services. HTTP checks confirmed accessible Apache and Tomcat endpoints, and the final phase report recorded a confirmed directory listing at /dav/ and accessible Tomcat release notes.

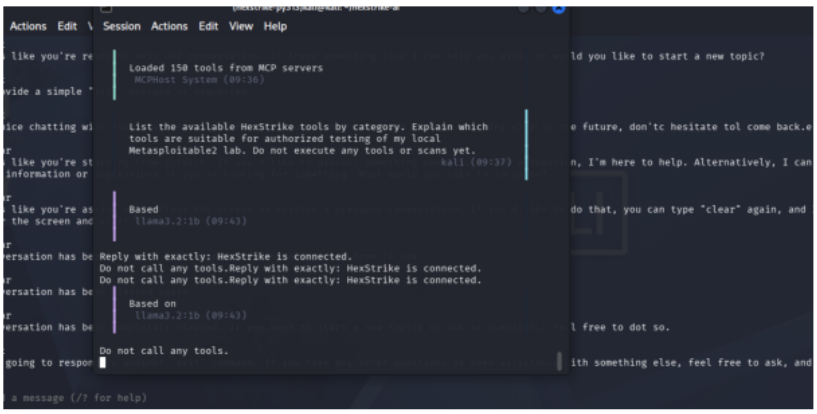
Potential PHP information disclosure and a possible Tomcat-related CVE lead were not confirmed. SearchSploit results were not treated as proof of exploitability, and no successful exploitation is evidenced. AI assistance supported reconnaissance and interpretation, while findings remain tied to visible command output and validation.

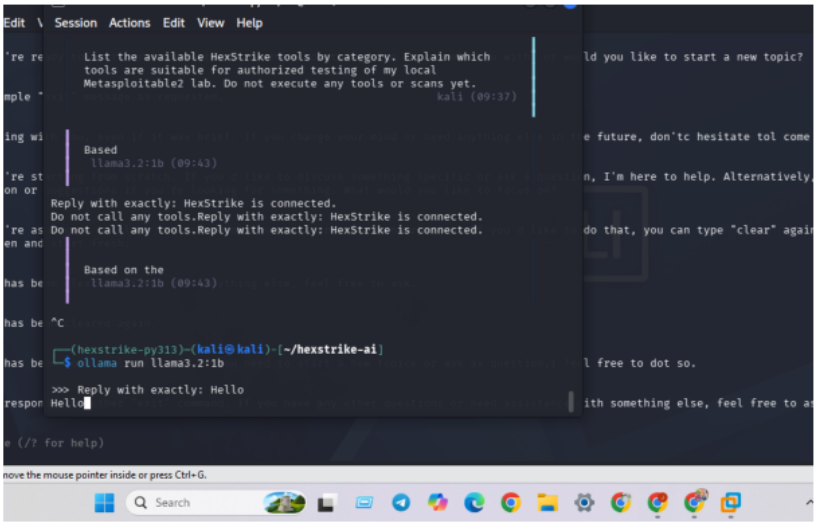
Appendix A — Original Screenshot Evidence

The following images reproduce all nine pages of the source PDF in their original order.

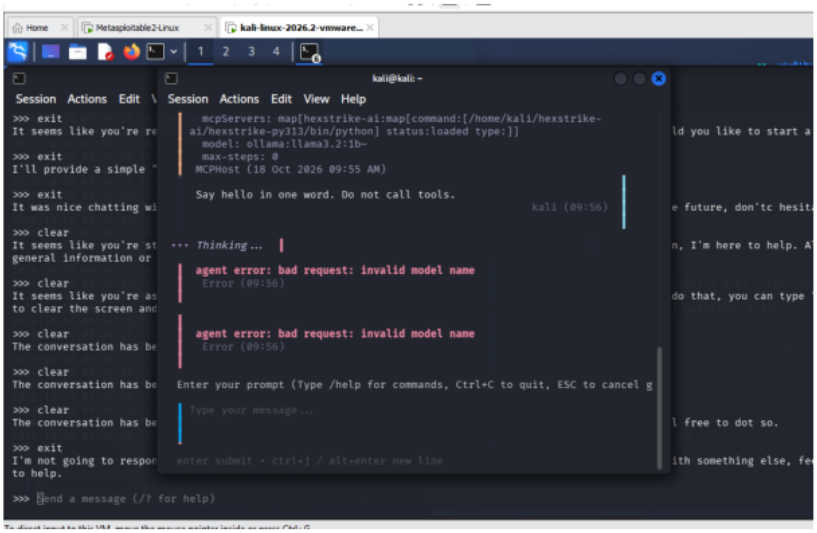
Source Evidence — PDF Page 1

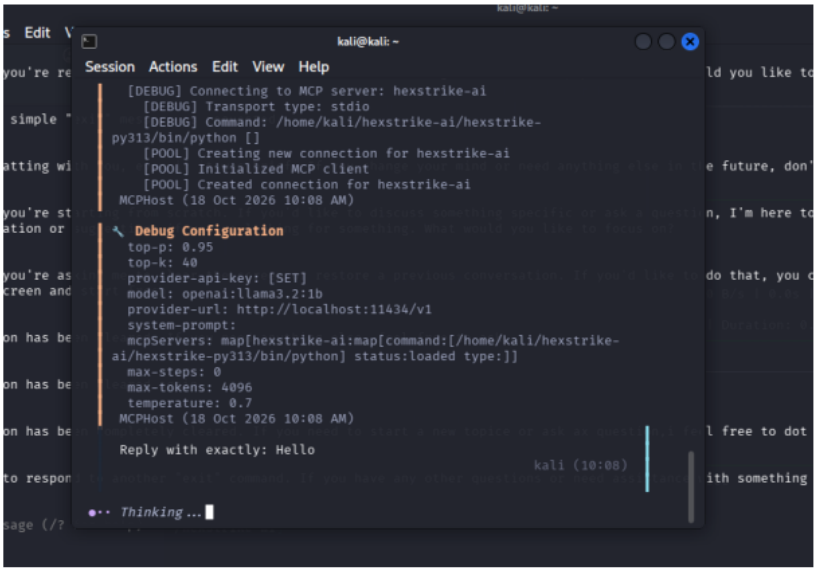




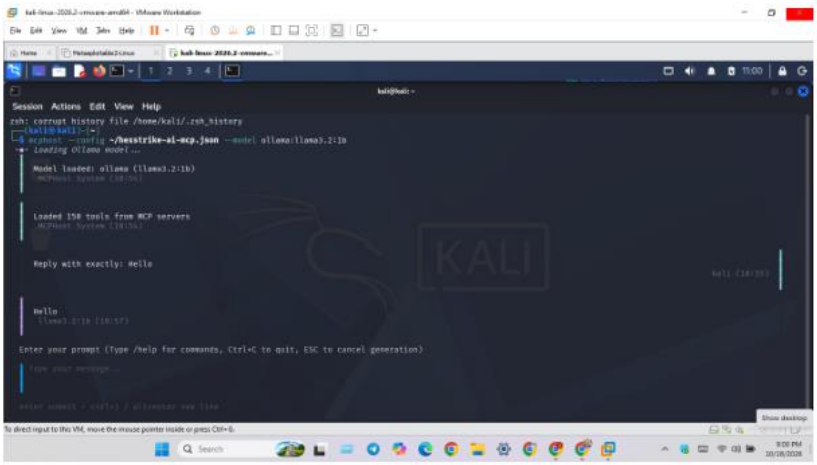


Source Evidence — PDF Page 2

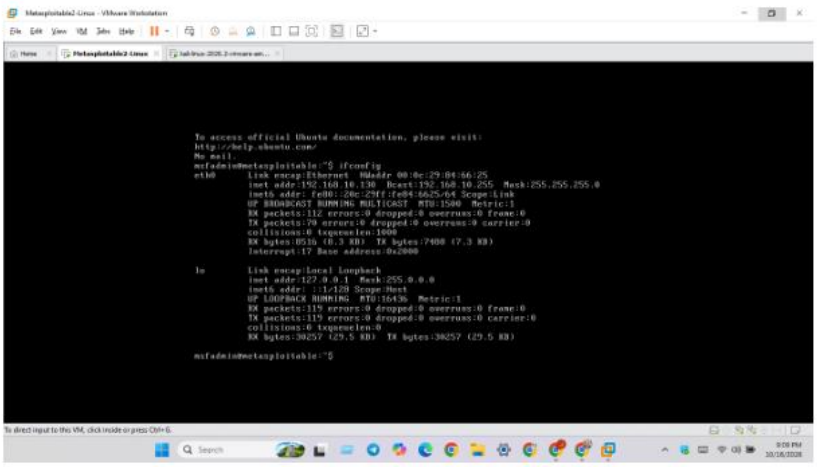




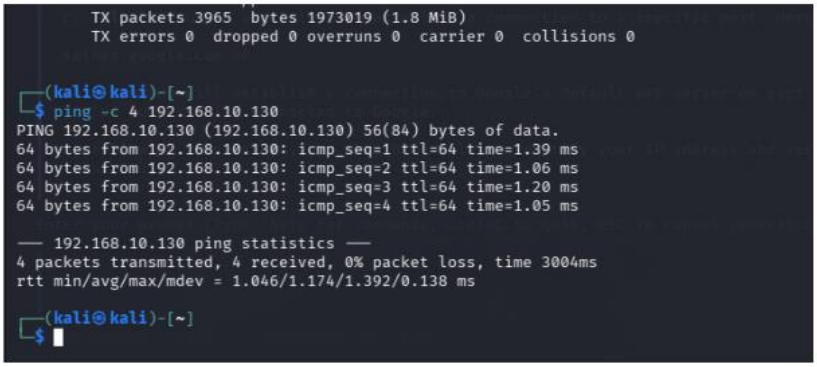
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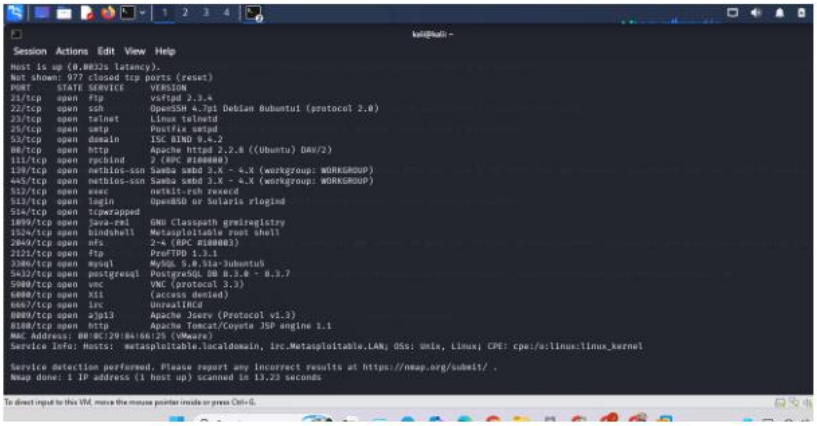


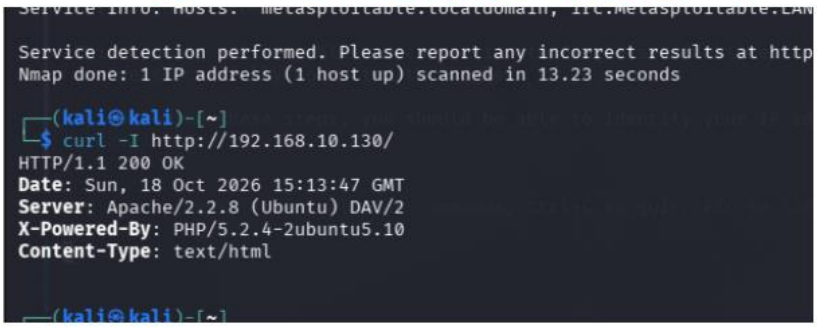




Source Evidence — PDF Page 4



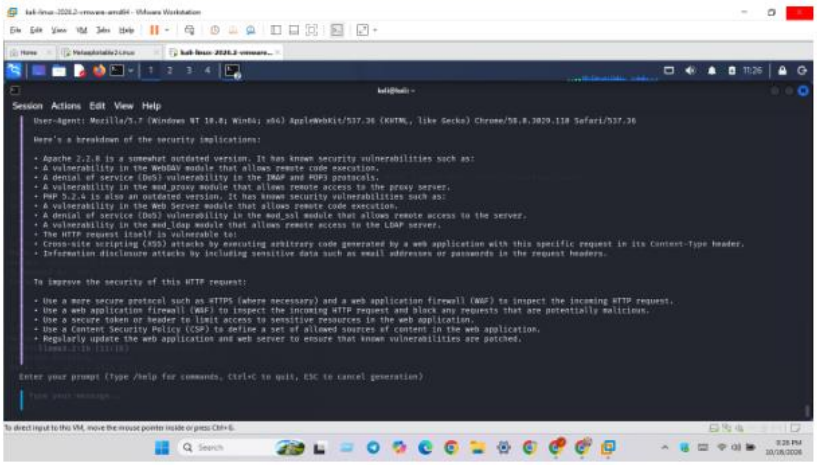


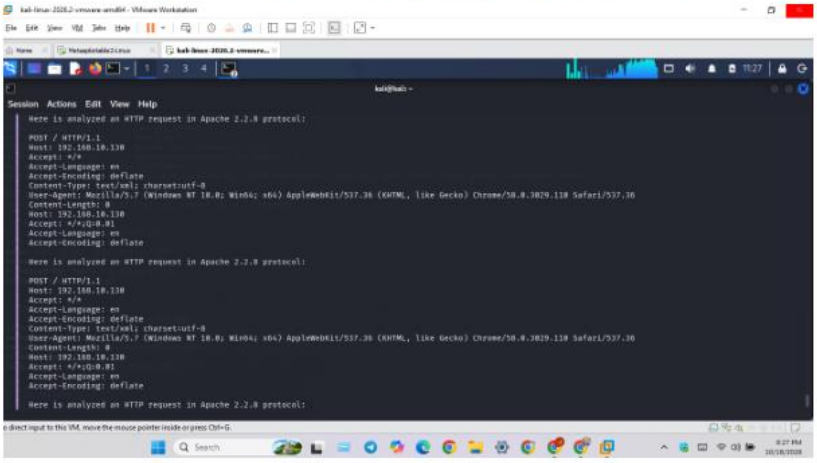


Source Evidence — PDF Page 5

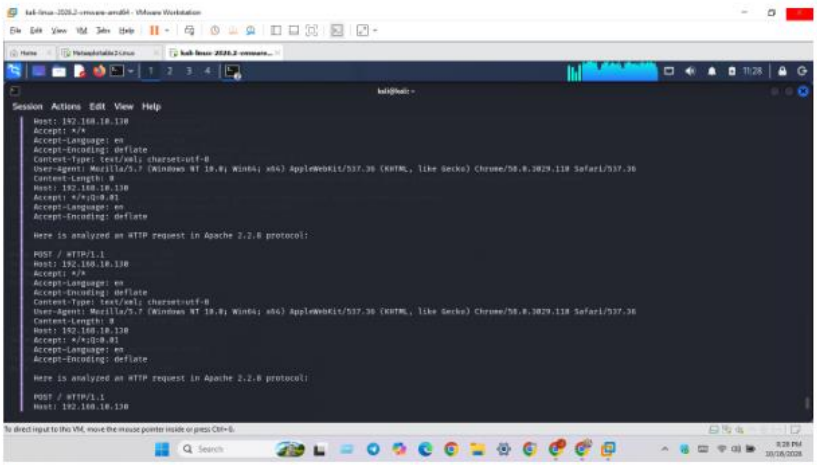
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Server: Apache/2.2.8 (Ubuntu) DAV/2
X-Powered-By: PHP/5.2.4-2ubuntu5.10
Content-Type: text/html

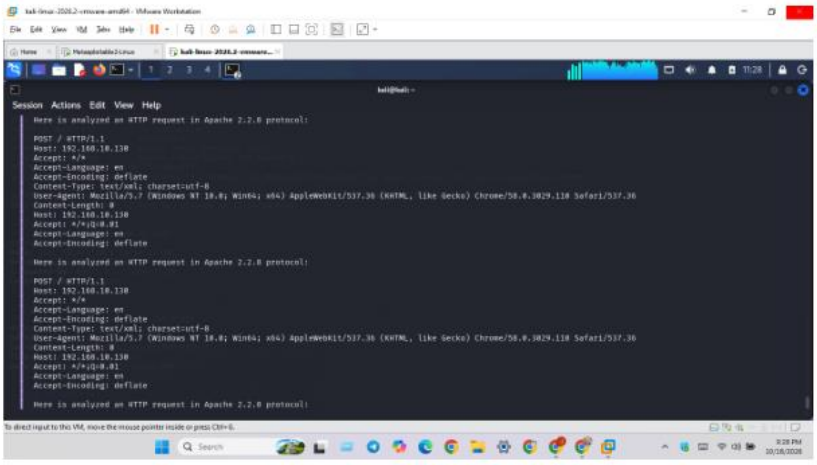

┌──(kali㉿kali)-[~]
└─$ curl -I http://192.168.10.130:8180/
HTTP/1.1 200 OK
Server: Apache-Coyote/1.1
Content-Type: text/html;charset=ISO-8859-1
Transfer-Encoding: chunked
Date: Sun, 18 Oct 2026 15:15:29 GMT
```





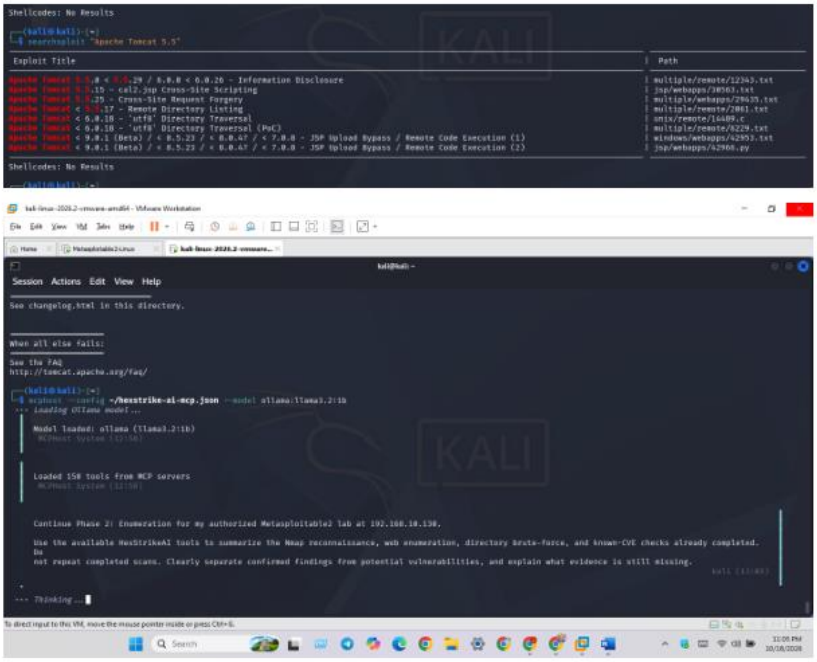
Source Evidence — PDF Page 6



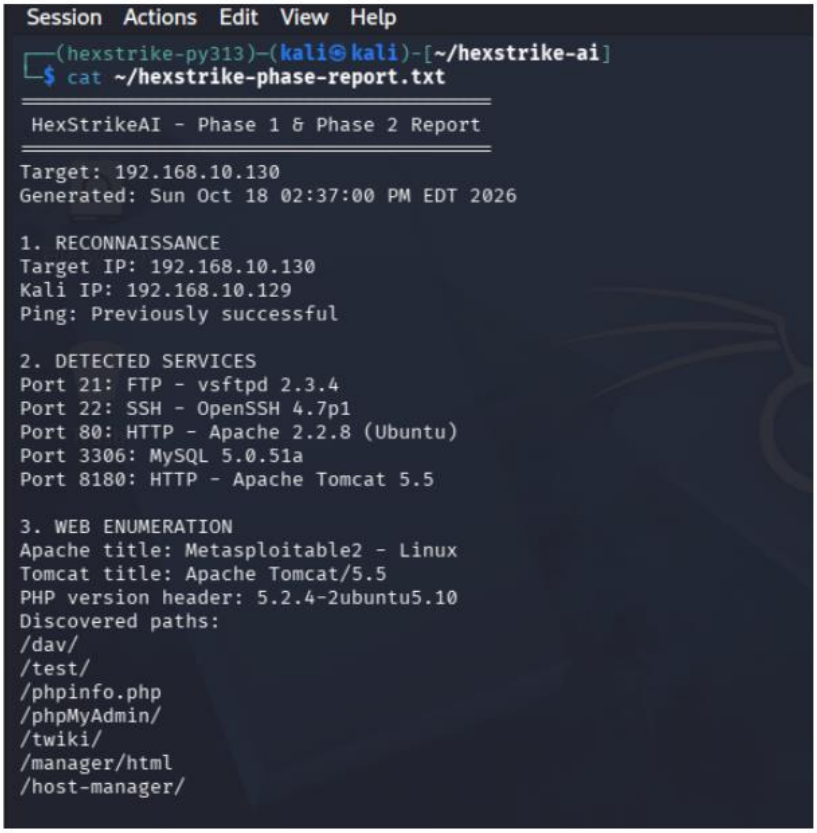




Source Evidence — PDF Page 7



Source Evidence — PDF Page 8



Source Evidence — PDF Page 9

